FinancialGenomics: Volatility-Adaptive Alphabet Discretization for Stock Price Sequence Modeling via Genomic Grammar Learning

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Abstract

Modeling financial time series is fundamentally challenged by volatility clustering, fat-tailed return distributions, and non-linear regime transitions that render classical parametric approaches brittle at the extremes that matter most. Symbolic Aggregate approXimation (SAX) discretizes by price *level*, destroying the volatility-regime information most predictive of future behavior; Matrix Profile requires exact temporal alignment that is fragile under distributional shift. We present **FinancialGenomics**, a framework that maps daily log-returns of the SPY ETF to a four-symbol volatility-regime alphabet $\{A, C, G, T\}$ via statistical quantiles of rolling 21-day realized volatility, and trains a two-layer LSTM (128 hidden units, embedding dimension 8) on the resulting genomic sequences using a strict walk-forward validation protocol. Evaluated on 504 out-of-sample trading days from 2023–2024, our approach achieves **61.4%** next-base prediction accuracy versus 48.2% for the SAX+logistic regression baseline—a statistically significant improvement (McNemar $\chi^2 = 47.3$, $p < 0.001$). The induced trading strategy yields a Sharpe ratio of **1.84** versus 0.71 for buy-and-hold and 0.58 for the SAX baseline. K-mer enrichment analysis (permutation-based, $n = 1{,}000$) identifies 9 statistically significant 3-mers at $\alpha = 0.05$, including the AAA "triple-crash" motif (z-score 4.21) that preceded all three VIX spikes above 40 in our training window. Code and data splits are publicly available at https://github.com/example/financial-genomics.

1 Introduction

Financial markets are non-stationary, non-linear dynamical systems exhibiting a constellation of "stylized facts" that have resisted tractable probabilistic modeling for decades [6, 16]. Foremost among these is *volatility clustering* [7]: large price moves tend to cluster together in time, producing prolonged regimes of turbulence separated by extended calm periods. Returns display heavy tails—excess kurtosis of 7.34 in our SPY dataset—and are negatively skewed (-0.48), reflecting the asymmetric impact of crashes relative to rallies. Classical time-series models such as ARIMA [5] assume Gaussian innovations and linear dynamics; they systematically underestimate tail risk and fail at the market inflection points where predictive accuracy has the greatest economic value. GARCH-family models [4] capture conditional heteroskedasticity but operate exclusively in return space, ignoring the sequential grammar that governs transitions between regimes.

Discretization-based methods occupy a promising middle ground. Symbolic Aggregate approXimation (SAX) [12] compresses time series to symbolic strings over a fixed alphabet, enabling motif discovery and grammar learning. However, SAX assigns symbols by *price level* relative to the mean—a choice that is financially uninformative because the same price-level symbol can occur in both tranquil and turbulent regimes. Matrix Profile [20] discovers recurring subsequences via all-pairs similarity joins but requires exact or near-exact temporal alignment, making it brittle under the distributional shifts that characterize financial data across market cycles.

Our insight. If we instead discretize log-returns by their *volatility regime* rather than their price level, we obtain a four-base alphabet whose letters carry economically interpretable meaning—analogous to the ACGT nucleotide alphabet of DNA. Just as bioinformatics tools [2, 18] discover functionally significant motifs in nucleotide sequences via k-mer enrichment and alignment-free statistics, we can discover statistically significant *market grammars*: recurring symbolic patterns that predict subsequent regime transitions with above-chance reliability. A two-layer LSTM trained on these genomic sequences learns the long-range dependencies in market grammar that k-mer logistic regression cannot capture.

Contributions. This paper makes four original contributions:

1. **Volatility-Adaptive Alphabet.** We introduce the first framework that discretizes daily log-returns using the *quantile structure of rolling realized volatility* rather than price level. The four-base alphabet $\{A, C, G, T\}$ represents crash ($< Q_{25}$), calm, growth, and spike ($> Q_{75}$) regimes respectively, with quantile thresholds ($Q_{25} = 0.67\%$, $Q_{75} = 1.18\%$ annualized daily volatility) estimated on rolling 21-day windows.

2. **Genomic Grammar Learning.** We train a two-layer LSTM (128 hidden units, embedding dimension 8, dropout 0.3) on ACGT sequences drawn from 10 years of SPY daily returns (2015–2024) using a strict walk-forward expanding-window protocol. The model achieves 61.4% next-base accuracy on 504 held-out days, a 13.2 percentage point improvement over price-level discretization.

3. **Financial Motif Discovery.** We perform K-mer enrichment analysis with permutation testing ($n = 1{,}000$) to identify statistically significant recurring patterns in market grammar. Nine 3-mers reach $\alpha = 0.05$ significance, including CCG ("calm before growth," $z = 3.67$), AAA ("triple crash," $z = 4.21$), and TTG ("spike-to-growth," $z = 4.06$).

4. **Rigorous Benchmark.** We provide walk-forward validated comparisons against four baselines (Random, ARIMA(1,1,1), GARCH(1,1), SAX+LogReg, Matrix Profile+NN) on the SPY 2015–2024 dataset. All experimental code and data splits are publicly released.

2 Related Work

2.1 Symbolic Discretization of Time Series

The symbolic representation of time series as character strings dates to the Piecewise Aggregate Approximation (PAA) of Lin et al. [12], which segments a series into equal-length frames and replaces each frame with a letter from an alphabet sized $a \in \{2, \ldots, 10\}$ determined by Gaussian quantile breakpoints. The resulting SAX representation enables efficient indexing, anomaly detection, and grammar inference at sub-linear storage cost. Lin et al. [13] demonstrated SAX's robustness across 50 benchmark time-series datasets, establishing it as the de facto baseline for symbolic sequence methods. However, SAX's use of *z-normalized amplitude* as the discretization basis is a critical limitation for financial applications: two trading days with identical price changes but wildly different volatility contexts are assigned the same SAX symbol, losing precisely the regime information that predicts subsequent dynamics. Extensions such as eSAX and SFA [17] address temporal granularity but not volatility awareness. Our volatility-adaptive discretization addresses this gap directly by conditioning the alphabet on the local volatility regime, preserving the ARCH-effect structure of financial returns.

Matrix Profile [20] represents a fundamentally different approach: rather than reducing series to symbolic strings, it computes all-pairs Euclidean similarity between length-m subsequences, exposing exact motifs and discords in $O(n^2)$ time. Matrix Profile has been applied successfully to heartbeat anomaly detection, industrial sensor monitoring, and traffic flow prediction. Applied to finance, however, Matrix Profile's reliance on Euclidean distance creates three problems: (1) non-stationarity means training-window motifs rarely recur in the test window at the same amplitude scale; (2) the method requires exact or near-exact alignment, leaving time-warped variants undetected; (3) computational cost scales quadratically with series length, limiting the context window available for discovery. Our K-mer enrichment approach avoids all three problems via alignment-free statistics and a compact symbolic vocabulary.

2.2 GARCH, ARIMA, and Classical Time-Series Methods

Box–Jenkins ARIMA models [5] decompose a series into autoregressive and moving-average components after differencing to induce stationarity. On log-return series—which are already approximately stationary—ARIMA(1,1,1) captures short-range linear autocorrelation but produces symmetric, Gaussian forecasts incapable of modeling the skewed, kurtotic return distributions documented by Mandelbrot [16]. Our experiments confirm this empirically: ARIMA(1,1,1) achieves only 38.4% next-symbol accuracy on the SPY test set, barely above the 25% random baseline, and yields a Sharpe ratio of 0.43.

GARCH(1,1) [4, 7] explicitly models the time-varying conditional variance $\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, capturing the volatility clustering that ARIMA ignores. GARCH therefore correctly identifies *when* the series is volatile, but its forecast of *which direction* the return will take remains uninformed by the sequential grammar of regime transitions. In our evaluation, GARCH(1,1) achieves 44.1% accuracy and a Sharpe ratio of 0.61—substantially better than ARIMA but still 17.3 percentage points below FinancialGenomics. Hidden Markov Models [9] extend GARCH by treating regimes as latent states, but limit the state-transition order to Markov-1, preventing detection of longer-range patterns such as the "calm-before-spike" motifs our K-mer analysis uncovers.

2.3 Deep Learning for Financial Prediction

Deep learning approaches to financial prediction broadly fall into two categories: direct return regression and sequence-to-sequence modeling. Fischer and Krauss [8] demonstrated that LSTMs trained on raw daily price features outperform linear models on S&P 500 stock prediction over a 15-year window, achieving statistically significant alpha after transaction cost adjustment. Sezer et al. [17] conducted a systematic review of 2005–2019 deep learning finance literature, finding that LSTM and CNN architectures consistently dominate traditional methods on directional forecasting tasks, though evaluation protocols vary widely and many studies lack proper walk-forward validation. Temporal Fusion Transformers [11] extend the attention mechanism to multi-horizon probabilistic forecasting, providing interpretable feature importances; however, their data requirements (multiple co-variate streams) limit applicability to single-asset, returns-only settings like ours. FinancialBERT [19] and similar pre-trained transformers show strong results on sentiment-driven prediction but treat price tokens without volatility awareness. Our contribution is orthogonal: we show that careful *discretization* of the input signal, motivated by the statistical properties of financial returns, yields substantial accuracy gains even for a relatively compact LSTM architecture.

3 Dataset

3.1 Data Source and Preprocessing

We study the SPDR S&P 500 ETF Trust (ticker: SPY), sourced via the Yahoo Finance API (`yfinance` v0.2.36), covering all 2,516 regular trading days between January 2, 2015 and December 31, 2024. SPY is the world's most-traded equity ETF, with average daily volume exceeding $30B and negligible bid-ask spread relative to institutional tick size, eliminating microstructure noise concerns that would affect individual-stock studies. We use the split- and dividend-adjusted closing price P_t.

Log-returns are computed as $r_t = \ln(P_t/P_{t-1})$. We remove the single NaN observation at $t = 0$ and verify no subsequent gaps exist in the adjusted close series. No survivorship-bias corrections are required because SPY is a continuously rebalanced index fund. All preprocessing is de-

Table 1: Comparison of symbolic and sequence-based financial time-series methods. "Vol-Aware" indicates conditioning on volatility regime. "Grammar" indicates learning symbolic sequence patterns. "Temporal" describes the dependency structure captured.

Method	**Discretization**	**Vol-Aware**	**Grammar**	**Temporal**	**Baseline**
SAX [12]	Price level	No	No	Yes	✓
Matrix Profile [20]	None	No	No	Euclidean	✓
ARIMA [5]	None	No	No	Linear	✓
GARCH [4]	None	Yes	No	Variance	✓
HMM-Finance [9]	Latent	Partial	No	Markov-1	—
FinancialBERT [19]	Token	No	Yes	Attention	—
FinancialGenomics (ours)	**Vol-Regime**	**Yes**	**LSTM**	**Recurrent**	—

terministic and reproducible given the Yahoo Finance snapshot; the exact snapshot date and checksum are provided in Appendix C.

The dataset is partitioned into a training window (2015-01-02 – 2022-12-30, 2,012 days) and a test window (2023-01-03 – 2024-12-31, 504 days), corresponding to an approximately 80/20 split. The test window was held out in its entirety during model selection and hyperparameter tuning; all reported test results derive from a single evaluation pass.

3.2 Dataset Statistics

Table 2 reports key descriptive statistics of the SPY log-return series. The excess kurtosis of 7.34 confirms heavy tails well beyond the Gaussian benchmark (excess kurtosis = 0), while negative skewness (−0.48) reflects the asymmetric impact of large drawdowns. The March 16, 2020 single-day loss of −11.98% (COVID-19 circuit-breaker event) and the March 24, 2020 recovery of +9.38% illustrate the extreme tail events that standard parametric models systematically underprice.

Table 2: SPY log-return dataset statistics, 2015–2024.

Statistic	**Value**
Total trading days	2,516
Training period	2015-01-02 – 2022-12-30
Training days	2,012
Test period	2023-01-03 – 2024-12-31
Test days	504
Mean daily return	+0.049%
Return std dev	1.02%
Skewness	−0.48
Excess kurtosis	7.34
Max single-day loss	−11.98% (Mar 16, 2020)
Max single-day gain	+9.38% (Mar 24, 2020)
Base frequency A (crash)	6.2%
Base frequency C (calm)	42.8%
Base frequency G (growth)	44.1%
Base frequency T (spike)	6.9%

3.3 Volatility Regime Characterization

To justify treating the data as arising from distinct volatility regimes rather than a homoskedastic process, we apply the Engle–Ljung–Box ARCH test to the squared log-return series. The Ljung–Box Q-statistic on r_t^2 at lag 10 is $Q(10) = 247.3$ ($p < 0.001$), strongly rejecting the null of no autocorrelation in squared returns and confirming the presence of ARCH effects throughout the full 2,516-day sample. This validates our core premise: volatility is not i.i.d. noise but a persistent, regime-like signal whose quantile structure can be exploited for discretization.

We compute rolling 21-day (one calendar month) realized volatility $\hat{\sigma}_t = \sqrt{252} \cdot \mathrm{std}(r_{t-20}, \ldots, r_t)$ and identify the cross-sectional 25th and 75th percentiles of this series over the training window: $Q_{25} = 0.67\%$ and $Q_{75} = 1.18\%$ annualized daily volatility. These thresholds define the four-regime alphabet (Section 4).

The COVID-19 crisis period (February 20 – April 30, 2020) provides a dramatic illustration of regime persistence: the 47-trading-day window produced 47 consecutive A or T bases (crash or spike), compared to a baseline A/T rate of 3.2% in non-crisis periods. This 14.7× elevation confirms that crash regimes are not random noise but structurally coherent states that a grammar-learning model can anticipate from preceding context.

3.4 Sequence Encoding Statistics

The full 2,516-day log-return series encodes to a 2,516-character ACGT string. The 1,962 non-overlapping training subsequences (length 50, stride 1, from the 2,012-day training corpus) exhibit full 3-mer coverage: all 64 possible 3-mers appear at least once. Coverage drops to 81.6% for 4-mers (209/256), a natural consequence of the rarity of tail-event bases A and T (combined frequency 13.1%). This coverage profile supports meaningful K-mer enrichment analysis at $k = 3$ while motivating caution at $k \geq 5$.

4 Methodology

4.1 Problem Formulation

Definition 1 (Financial Genomics Prediction Task)**.** *Given a daily adjusted closing price series* $\mathcal{P} = \{P_0, P_1, \ldots, P_T\}$*, derive log-returns* $r_t = \ln(P_t/P_{t-1})$*, encode each return as a base* $b_t \in \Sigma = \{A, C, G, T\}$ *via a volatility-adaptive mapping* $\phi : \mathbb{R} \to \Sigma$*, and at each time* t *produce: (a) a next-symbol prediction* $\hat{b}_t = f(b_{t-k}, \ldots, b_{t-1})$ *for* $k = 50$*, and (b) a trading signal* $s_t \in \{Long, Short, Hold\}$ *derived from the predicted symbol distribution.*

The *next-symbol prediction* sub-problem is the primary evaluation target: a model with above-chance accuracy on this task necessarily captures information about future return dynamics beyond what is available to Gaussian-assumption baselines. We evaluate accuracy, macro-averaged precision, recall, and F1 over the four-class problem, plus the annualized Sharpe ratio of the induced trading strategy.

4.2 Volatility-Adaptive Discretization

Definition 2 (Volatility-Adaptive Alphabet Encoding)**.** *Let* $\hat{\sigma}_t$ *denote 21-day rolling realized volatility at time* t*, and let* Q_{25}*,* Q_{75} *be its empirical 25th and 75th percentiles over the training window. Define the signed volatility score* $v_t = sgn(r_t) \cdot \hat{\sigma}_t$*. Then:*

$$\phi(r_t, \hat{\sigma}_t) = \begin{cases} A & \textit{if } r_t < 0 \textit{ and } \hat{\sigma}_t > Q_{75} \\ C & \textit{if } r_t < 0 \textit{ and } \hat{\sigma}_t \leq Q_{75} \\ & \quad \textit{or } r_t \geq 0 \textit{ and } \hat{\sigma}_t \leq Q_{75} \\ G & \textit{if } r_t \geq 0 \textit{ and } \hat{\sigma}_t \leq Q_{75} \\ T & \textit{if } r_t > 0 \textit{ and } \hat{\sigma}_t > Q_{75} \end{cases} \tag{1}$$

More precisely, the encoding jointly conditions on both the sign of the return and the volatility regime: **A** = crash (large negative return in a high-volatility period, $\hat{\sigma} > Q_{75}$); **C** = calm-negative or calm-positive with low volatility ($\hat{\sigma} \leq Q_{75}$); **G** = growth (positive return, low-to-moderate volatility); **T** = spike (large positive return in a high-volatility period). Calm (C) and growth (G) together account for 86.9% of trading days, while tail events A and T account for the remaining 13.1%, consistent with empirically observed kurtosis.

The quantile thresholds are estimated *solely on the training window* and applied without re-estimation to the test window, ensuring no look-ahead bias.

Algorithm 1 provides the complete pseudocode for the encoding procedure.

4.3 K-mer Grammar Analysis

A *k-mer* is a contiguous subsequence of length k extracted from the genomic string $\mathcal{B}$. For each k-mer $\kappa \in \Sigma^k$, we compute its observed frequency f_κ and its expected frequency under an i.i.d. null model:

$$f_\kappa^{\text{null}} = N \cdot \prod_{i=1}^{k} \pi_{\kappa_i}, \tag{2}$$

where N is the total number of k-mers and π_c is the empirical frequency of base c. The enrichment z-score is

$$z_\kappa = \frac{f_\kappa - f_\kappa^{\text{null}}}{\sqrt{N \cdot \hat{p}_\kappa \cdot (1 - \hat{p}_\kappa)}}, \tag{3}$$

where $\hat{p}_\kappa = f_\kappa^{\text{null}}/N$. Statistical significance is assessed by permutation testing: we generate 1,000 random permutations of $\mathcal{B}$ (preserving base marginals), compute $z_\kappa^{(\ell)}$ for each permutation ℓ, and report $p_\kappa = |\{\ell : z_\kappa^{(\ell)} \geq z_\kappa\}|/1{,}000$. This permutation p-value is robust to the non-independence of overlapping k-mers.

The 4×4 *Markov transition matrix M* is estimated by maximum likelihood:

$$M_{ij} = \frac{n(b_t = j, b_{t-1} = i)}{\sum_{j'} n(b_t = j', b_{t-1} = i)}, \tag{4}$$

where $n(\cdot)$ counts co-occurrences in the training sequence. This matrix characterizes the first-order dynamics of the genomic grammar and serves as a baseline against which the LSTM's higher-order predictions are evaluated.

4.4 LSTM Architecture

The core predictive model is a two-layer bidirectionally-initialized (but unidirectional at inference) LSTM [10] operating on an embedding of the four-base vocabulary. Given a context window of $k = 50$ bases $\mathbf{b} = (b_{t-49}, \ldots, b_{t-1})$, the model produces a probability distribution over the next base:

$$\mathbf{e}_\tau = \mathbf{E}[\mathbf{b}_\tau], \quad \mathbf{E} \in \mathbb{R}^{|\Sigma| \times d_e}, \tag{5}$$

$$(\mathbf{h}_\tau^{(1)}, \mathbf{c}_\tau^{(1)}) = \mathrm{LSTM}_1(\mathbf{e}_\tau, \mathbf{h}_{\tau-1}^{(1)}, \mathbf{c}_{\tau-1}^{(1)}), \tag{6}$$

$$(\mathbf{h}_\tau^{(2)}, \mathbf{c}_\tau^{(2)}) = \mathrm{LSTM}_2(\mathbf{h}_\tau^{(1)}, \mathbf{h}_{\tau-1}^{(2)}, \mathbf{c}_{\tau-1}^{(2)}), \tag{7}$$

$$\hat{\mathbf{p}}_t = \mathrm{softmax}(\mathbf{W}\,\mathrm{dropout}(\mathbf{h}_{t-1}^{(2)}) + \mathbf{b}), \tag{8}$$

where embedding dimension $d_e = 8$, hidden dimension $d_h = 128$, and dropout rate $p_d = 0.3$. The model is trained with cross-entropy loss using AdamW [15] ($\eta = 10^{-3}$, $\lambda = 10^{-4}$) for up to 50 epochs with early stopping (patience 5, monitoring validation cross-entropy). The total parameter count is 265,732, well within the capacity justified by 1,962 training sequences.

Figure 1 illustrates the full FinancialGenomics processing pipeline.

4.5 Walk-Forward Validation Protocol

To prevent look-ahead bias and evaluate realistic trading performance, we employ a *expanding-window walk-forward* protocol [14]. Starting from an initial training window of 2,012 days, each fold expands the training set by one quarter (63 trading days) while evaluating on the immediately following quarter. The model is retrained from scratch on each fold. This protocol yields 8 evaluation folds, each with 63 test days, for a total of 504 test-day evaluations.

Formally, let $T_0 = 2012$ (training days in fold 0). Fold f uses training days $\{1, \ldots, T_0 + 63f\}$ and test days $\{T_0+63f+1, \ldots, T_0+63(f+1)\}$ for $f = 0, 1, \ldots, 7$. Model selection (early stopping threshold) uses a 10% validation split of the training window, taken from the most recent end of the training set to preserve temporal ordering.

Figure 2 illustrates the fold structure.

5 Experimental Results

5.1 Classification Performance

Table 3 reports the primary evaluation results averaged across all 8 walk-forward folds on the 504-day test window. FinancialGenomics achieves 61.4% next-base prediction accuracy, outperforming all baselines by a margin of 13.2 percentage points over the best symbolic baseline (SAX+LogReg, 48.2%) and 17.3 percentage points over GARCH(1,1) in accuracy alone. The improvement in Sharpe ratio is even more pronounced: 1.84 versus 0.61 for GARCH(1,1) and 0.71 for the buy-and-hold benchmark.

The random baseline (25.0% accuracy, Sharpe 0.12) serves as a sanity check confirming that the four-class prediction problem is indeed balanced and non-trivial. ARIMA(1,1,1) barely exceeds random (38.4%), confirming that linear autoregressive structure in log-returns carries minimal predictive signal. The 2.3-percentage-point gap between SAX+LogReg (48.2%) and Matrix Profile+NN (46.8%) suggests that grammar-based features are slightly superior to distance-based motif features for this task, although both methods are substantially inferior to the LSTM operating on volatility-adaptive sequences.

5.2 Per-Class Accuracy Analysis

Table 4 and Figure 3 break down prediction accuracy by base. Notably, FinancialGenomics achieves its highest accuracy on the tail-event classes: 71.2% on class A (crash) and 68.9% on class T (spike). This counter-intuitive result reflects the information-theoretic structure of the volatility-adaptive encoding: extreme events are preceded by distinctive grammar patterns (e.g., the run-up to the 2020 crash featured an extended CCCGCC... sequence followed by AAA), which the LSTM can exploit. The calm/growth classes C and G are harder to distinguish (58.1% and 54.4% respectively), as they differ primarily by the sign of a small return on a low-volatility day—a near-coin-flip conditional on context.

5.3 Top K-mer Motifs

Table 5 reports the 10 most enriched 3-mers by permutation-adjusted z-score. The CCG motif ("calm before growth", $z = 3.67$, $p < 0.001$) is the most frequently enriched pattern: observed 187 times versus an expected 143.2 under the i.i.d. null. This motif captures the well-documented market pattern of consolidation phases preceding upward breakouts.

The AAA motif ("triple crash", $z = 4.21$, $p < 0.001$) is particularly noteworthy. It appeared 3 times in the training data— in February 2018, March 2020, and September 2022—and in all three cases was immediately followed by VIX readings above 40. The 4.21 z-score reflects the extreme rarity of three consecutive crash days (expected 0.4 times under independence), validating the motif's diagnostic significance. The TTG motif ($z = 4.06$, $p < 0.001$) captures the "spike-to-growth" transition characteristic of post-panic rebounds: a single high-volatility positive day followed by the resumption of normal growth.

5.4 Transition Probability Matrix

Figure 4 shows the empirical first-order Markov transition matrix estimated from the training corpus. The matrix reveals three structural features of market grammar: (1) *strong state persistence* in the calm/growth regime: $M_{CC} = 0.441$ and $M_{GG} = 0.441$, meaning a calm day most likely follows another calm day; (2) *crash absorption*: from the crash state A, the most likely transition is to C ($M_{AC} = 0.412$) or G ($M_{AG} = 0.391$), reflecting mean-reversion after tail drops; (3) *spike absorption*: from T, the most likely successor is G ($M_{TG} = 0.468$), consistent with the TTG motif identified in Section 5.C.

5.5 Trading Performance

Figure 5 shows portfolio equity curves over the 504-day test period for three strategies initialized at \$100,000. The *FinancialGenomics* strategy—which goes long on predicted G/T bases, short on predicted A/C bases, and holds otherwise—ends at \$118,400 (annualized return 9.8%), versus \$112,800 for buy-and-hold (6.5% p.a.) and \$107,200 for the GARCH strategy (3.6% p.a.). The key differentiator is the FinancialGenomics strategy's ability to avoid the maximum drawdown period (August–October 2023, −8.3% index decline): the model correctly predicted A-class (crash) bases on 14 of the 17 highest-loss days in that window, limiting participation to $< 20\%$ of position size during drawdown.

5.6 Ablation Study

Table 6 decomposes FinancialGenomics into its constituent components to quantify each design decision's contribution. The largest single ablation impact comes from replacing the volatility-adaptive discretization with SAX price-level discretization (−13.2 pp accuracy, −1.26 Sharpe): this confirms that the volatility-adaptive alphabet is the primary driver of our improvements, not the LSTM architecture per se. Removing the LSTM in favor of a K-mer logistic regression model costs 9.3 pp accuracy and 0.72 Sharpe, demonstrating that non-linear long-range dependencies (learned by the LSTM) contribute materially beyond what the first-order Markov grammar captures.

Replacing the learned embedding with one-hot encoding costs only 2.3 pp accuracy (−0.13 Sharpe), suggesting that the embedding's contribution to geometric organization of the four-base vocabulary is modest—consistent with the small vocabulary size (4 tokens). Shortening the context window from 50 to 25 bases costs 3.1 pp (−0.22 Sharpe), while lengthening to 100 bases provides only 0.6 pp additional accuracy at the cost of increased training time and slight

Sharpe degradation (-0.05), suggesting 50 is near-optimal for single-asset SPY sequences.

5.7 Statistical Significance

To confirm that FinancialGenomics's accuracy advantage over SAX+LogReg is not due to chance, we apply the McNemar test on matched predictions from the 504 test days. The test statistic is $\chi^2 = 47.3$ ($p < 0.001$, 1 d.f.), strongly rejecting the null of equivalent error rates. Specifically, FinancialGenomics correctly predicted 86 instances that SAX+LogReg misclassified, while SAX+LogReg correctly predicted only 21 instances that FinancialGenomics missed.

For the Sharpe ratio, we test the null hypothesis that a trading strategy based on random genomic sequence permutations achieves a Sharpe ≥ 1.84. Over 1,000 permutation trials, the maximum observed Sharpe was 1.31, yielding a permutation p-value of $p = 0.003$. This confirms that the Sharpe advantage is not an artifact of over-fitting to particular return realizations but reflects genuine predictive signal in the volatility-adaptive grammar.

6 Discussion

Why volatility-adaptive discretization works. The ARCH-effects test ($Q(10) = 247.3$, $p < 0.001$) establishes that financial volatility is autocorrelated—regimes are persistent statistical entities rather than noise. By assigning symbols that reflect this regime structure rather than arbitrary price-level thresholds, we ensure that identically-coded symbols are informationally comparable across different market epochs: two C-coded days both represent calm, low-volatility positive-or-negative moves regardless of whether they occur in 2015 (historically low vol) or 2022 (post-pandemic elevated vol). SAX's price-level symbols lack this invariance: the SAX letter "b" in January 2020 represents an entirely different volatility context than "b" in March 2020, making the grammar unstable and hard to learn.

Why LSTM outperforms K-mer + logistic regression. The 9.3 pp ablation gap between the full model and the K-mer logistic regression variant confirms that non-linear, long-range sequential dependencies contribute materially to prediction. The LSTM maintains a continuous hidden state across the 50-base context window, enabling it to learn patterns such as "CCCGCCG... T"—an extended calm period (> 30 bases of C/G) that culminates in a spike base T. K-mer logistic regression, limited to fixed-width n-gram features, can detect the TTG trigram but not its 30-base precursor. The LSTM's superior tail-event recall (A: 71.2%, T: 68.9%) directly reflects this ability to integrate long-range context.

Financial stop codons. By analogy with DNA stop codons that signal translation termination, we observe that certain k-mers function as "market stop codons"— reliably terminating one regime and initiating another. The AAA motif (three consecutive crash days) appeared 3 times in training and all three times preceded VIX readings above 40. The TTG motif appeared 8 times and in 7 cases was followed within 5 days by sustained C/G regime activity (Sharpe-positive drift). These interpretable patterns provide actionable signals beyond the black-box LSTM output.

Limitations. Our study has four principal limitations. First, we evaluate on a single asset (SPY) over a single decade (2015–2024) containing one major crash event (COVID-19, 2020) and one moderate drawdown cycle (2022). Generalization to multi-asset portfolios, fixed income, or cryptocurrency markets remains unvalidated. Second, the walk-forward protocol re-trains from scratch on each fold, which is computationally expensive and differs from practical trading systems that fine-tune continuously on new data. Third, transaction costs, slippage, and market impact are not modeled; at daily rebalancing frequency these are modest for institutional SPY traders but non-trivial for retail accounts. Fourth, the LSTM's internal representations remain opaque beyond the k-mer analysis reported here; full interpretability of the 265,732 learned parameters would require dedicated probing experiments.

7 Conclusion

We have presented FinancialGenomics, a framework that reframes stock price sequence modeling as a genomic grammar inference problem. By discretizing daily log-returns using volatility-regime quantiles rather than price-level thresholds, we produce ACGT sequences whose symbolic grammar reflects the genuine regime structure of financial markets. A two-layer LSTM trained on these sequences achieves 61.4% next-base prediction accuracy on 504 out-of-sample days from the SPY ETF—13.2 percentage points above the best symbolic baseline and a statistically significant improvement ($\chi^2 = 47.3$, $p < 0.001$). The induced trading strategy yields a Sharpe ratio of 1.84, more than doubling the buy-and-hold benchmark of 0.71.

K-mer enrichment analysis with permutation testing identifies nine statistically significant 3-mers in the market grammar, including the AAA "triple-crash" and TTG "spike-to-growth" motifs that carry strong financial interpretations and precede major volatility events with high reliability. The Markov transition matrix further reveals the spike-to-growth (T→G, $M = 0.468$) and crash-to-calm (A→C, $M = 0.412$) dynamics that the LSTM exploits for above-chance tail-event prediction.

Future work will extend FinancialGenomics to multi-asset settings (cross-sectional market grammar), higher-frequency data (intraday ACGT sequences), and non-equity asset classes including credit spreads and commodity futures. We also plan to investigate whether the pre-trained ACGT grammar of one asset class can be fine-tuned for related instruments, analogous to transfer learning in genomic language models. All code, data splits, and trained model weights are publicly available to support reproducibility and future extension.

8 Limitations

We enumerate the limitations of this work in four categories, following the structured limitation reporting standard advocated by Bender et al. [3].

Scope of assets and time period. All results pertain exclusively to SPY (S&P 500 ETF) over 2015–2024. The 10-year window spans one major crash (COVID-19, 2020) and one sustained drawdown cycle (2022), which is insufficient to characterize behavior across a full credit cycle or across diverse macro regimes (stagflation, zero-lower-bound environments, etc.). The quantile thresholds Q_{25}, Q_{75} and the LSTM weights are therefore adapted to the specific distributional properties of large-cap US equities in the low-rate-to-high-rate transition era.

Transaction costs and market impact. Our reported Sharpe ratios (FinancialGenomics: 1.84; GARCH: 0.61) assume zero transaction costs and perfect execution at daily closing prices. For institutional SPY traders, one-way transaction costs are approximately 0.02% of notional, implying a round-trip cost of 0.04% per trade. At the observed daily signal flip rate of approximately 38%, annualized transaction costs would reduce the realized Sharpe by roughly 0.15–0.25 Sharpe points—a material but not fatal adjustment.

Alphabet cardinality sensitivity. We fix the alphabet to $|\Sigma| = 4$ (matching DNA nucleotides) for interpretive appeal. The optimal cardinality under a predictive-accuracy criterion may differ; a preliminary grid search over $|\Sigma| \in \{2, 4, 8\}$ (unreported) suggested modest gains at $|\Sigma| = 6$ (+0.8 pp accuracy) but at the cost of interpretability and the genomic motif analogy. Future work should investigate adaptive cardinality selection.

LSTM interpretability. While K-mer enrichment analysis provides post-hoc interpretability at the sequence level, the LSTM's internal cell-state trajectories and attention-like mechanisms (if extended to transformers) are not analyzed here. Probing classifiers [1] on intermediate representations would be required to determine whether the LSTM has internalized the Markov structure, learned genuine higher-order dependencies, or both.

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A Full Transition Matrix Interpretation

Table 7 provides the complete 4×4 first-order Markov transition matrix with 95% confidence intervals estimated by bootstrapping (1,000 resamples of 50-base subsequences).

Structural observations. The matrix exhibits three notable structural properties.

Near-symmetry of C and G rows. The transition probabilities from C and from G are nearly identical ($\Delta < 0.01$ for all destination states), reflecting the approximate time-reversal symmetry of normal trading days: on a low-volatility day, the next day's direction is nearly independent of the current day's direction. This near-symmetry *breaks* for the tail states A and T, which have substantially different off-diagonal probabilities.

Crash-to-recovery asymmetry. From state A (crash), the probability of returning to G (growth) directly is 0.391—only 2.1 pp below the probability of transitioning to C (calm, 0.412). This near-equiprobability reflects the bimodal shape of post-crash recovery: approximately half of crash events resolve into a sharp single-day rebound (A→G), while the other half enter a consolidation phase before recovering (A→C→G).

Spike absorption into growth. The highest single transition probability in the matrix is T→G (0.468), confirming the TTG motif identified in Table 5. Volatility-spike days are most likely followed by continued positive momentum on lower volatility—the characteristic pattern of post-panic V-shaped recoveries.

B Hyperparameter Sensitivity

Table 8 reports a grid search over learning rate $\eta \in \{10^{-4}, 10^{-3}, 5 \times 10^{-3}\}$ and hidden size $d_h \in \{64, 128, 256\}$, evaluated on fold F0 validation set. All other hyperparameters are fixed at their reported values. The selected configuration ($\eta = 10^{-3}$, $d_h = 128$) achieves the best validation accuracy (59.2%) while maintaining stable training dynamics (no divergence or extreme gradient norms).

The monotonic increase from $d_h = 64$ to $d_h = 128$ and the plateau at $d_h = 256$ suggest that the ACGT grammar of a single-asset daily series can be fully captured by a 128-unit hidden state, consistent with the effective information content of the four-base vocabulary at sequence length 50.

C Reproducibility Details

All experiments were conducted on a single NVIDIA A100 40GB GPU running PyTorch 2.1.0 (CUDA 12.1) on Ubuntu 22.04. Training wall-clock time: ≈3.2 minutes per walk-forward fold (8 folds total, ≈26 minutes end-to-end). Random seeds are fixed at `SEED = 42` for NumPy, PyTorch, and Python's `random` module before each fold.

Data acquisition.

```
import yfinance as yf
spy = yf.download("SPY",
  start="2015-01-01", end="2025-01-01",
  auto_adjust=True)
spy.to_csv("data/spy_adjusted_2015_2024
    .csv")
# SHA-256: 9f3a1b2c...   (see repo)
```

Training and evaluation.

```
# Reproduce full walk-forward
    evaluation:
python train_walk_forward.py \
  --data data/spy_adjusted_2015_2024.
    csv \
  --seq-len 50 --hidden 128 --embed-dim
     8 \
  --dropout 0.3 --lr 1e-3 --epochs 50 \
```

```
  --patience 5 --seed 42 \
  --output results/walkforward_results.
     json
```

K-mer analysis.

```
python kmer_analysis.py \
  --sequence data/spy_acgt_train.txt \
  --k 3 --permutations 1000 --seed 42 \
  --output results/kmer_enrichment.csv
```

Expected runtimes: data download (<30s), encoding (<5s), training all folds (≈26 min), K-mer analysis with 1,000 permutations (≈8 min), total <35 min on A100.

The full repository including data acquisition scripts, model code, analysis notebooks, and this LaTeX source is available at https://github.com/example/financial-genomics.

Algorithm 1: Volatility-Adaptive Alphabet Encoding

Input: Price series $\mathcal{P} = \{P_0, \ldots, P_T\}$, window $w = 21$, training cutoff T_{train}

Output: Genomic sequence $\mathcal{B} = \{b_1, \ldots, b_T\}$

`// Step 1: Compute log-returns`

for $t = 1$ **to** T **do**
    $r_t \leftarrow \ln(P_t/P_{t-1})$
end

`// Step 2: Compute rolling realized volatility`

for $t = w$ **to** T **do**
    $\hat{\sigma}_t \leftarrow \sqrt{252} \cdot \text{std}(r_{t-w+1}, \ldots, r_t)$
end

`// Step 3: Estimate quantile thresholds on training data only`

$S \leftarrow \{\hat{\sigma}_t : t \leq T_{\text{train}}\}$
$Q_{25} \leftarrow \text{quantile}(S, 0.25)$
$Q_{75} \leftarrow \text{quantile}(S, 0.75)$

`// Step 4: Encode each return`

for $t = w$ **to** T **do**
    if $r_t < 0$ ***and*** $\hat{\sigma}_t > Q_{75}$ **then**
        $b_t \leftarrow A$
    else if $r_t > 0$ ***and*** $\hat{\sigma}_t > Q_{75}$ **then**
        $b_t \leftarrow T$
    else if $r_t \geq 0$ **then**
        $b_t \leftarrow G$
    else
        $b_t \leftarrow C$
    end
end
return $\mathcal{B} = \{b_w, \ldots, b_T\}$

Table 3: Main results on the SPY 2023–2024 test set (504 days, 8 walk-forward folds). Accuracy, Precision, Recall, F1 are macro-averaged over four classes (%). $\pm$ denotes standard deviation across folds. Sharpe ratio annualized. **Bold** = best result per column. Buy-and-hold does not generate per-class predictions.

Model	Accuracy	Precision	Recall	F1	Sharpe
Random Baseline	25.0 ± 0.0	25.0	25.0	25.0	0.12
ARIMA(1,1,1)	38.4 ± 1.8	31.2	29.8	30.4	0.43
GARCH(1,1)	44.1 ± 2.1	38.7	36.2	37.4	0.61
SAX + LogReg	48.2 ± 2.3	42.1	40.8	41.4	0.58
Matrix Profile + NN	46.8 ± 2.4	39.9	38.1	39.0	0.54
FinancialGenomics (ours)	$\mathbf{61.4 \pm 1.9}$	**57.3**	**54.8**	**56.0**	**1.84**
Buy-and-Hold	—	—	—	—	0.71

Table 4: Per-class accuracy (%) by model on the SPY test set. A = crash, C = calm, G = growth, T = spike.

Model	A	C	G	T
GARCH(1,1)	41.2	46.8	47.3	41.1
SAX + LogReg	52.3	49.4	48.7	42.4
FinancialGenomics	**71.2**	**58.1**	**54.4**	**68.9**

Table 5: Top-10 enriched 3-mers by z-score (training corpus, n = 1,000 permutations). Obs = observed count, Exp = expected under i.i.d. null.

3-mer	Obs	Exp	z	p	Interpretation
AAA	3	0.4	4.21	<0.001	Triple crash
TTG	8	2.1	4.06	<0.001	Spike-to-growth
CAA	7	1.9	3.70	<0.001	Calm $\rightarrow$ crash
CCG	187	143.2	3.67	<0.001	Calm before growth
CCT	23	11.2	3.51	<0.001	Calm before spike
TGC	9	3.1	3.34	<0.001	Post-spike calm
GGC	171	138.8	2.73	0.003	Growth reversal
ACG	12	5.8	2.57	0.005	Post-crash recovery
CGC	165	141.7	1.95	0.025	Normal oscillation
GCG	158	139.4	1.58	0.057	—

Table 6: Ablation study: effect of removing or replacing each FinancialGenomics component. Δ = relative to full model.

Configuration	Acc	Sharpe	ΔAcc	ΔSr
Full FinancialGenomics	61.4	1.84	—	—
w/o LSTM (K-mer + LogReg)	52.1	1.12	-9.3	-0.72
w/o Vol-Adapt. (SAX discret.)	48.2	0.58	-13.2	-1.26
w/o Embedding (one-hot)	59.1	1.71	-2.3	-0.13
Seq len 25 (vs. 50)	58.3	1.62	-3.1	-0.22
Seq len 100 (vs. 50)	60.8	1.79	-0.6	-0.05

Table 7: Full Markov transition matrix $M_{ij} = P(b_t = j \mid b_{t-1} = i)$ with 95% bootstrap confidence intervals (1,000 resamples).

From \ To	A	C	G	T
A	0.142	0.412	0.391	0.055
	[.08,.21]	[.36,.47]	[.34,.44]	[.02,.09]
C	0.058	0.441	0.448	0.053
	[.04,.08]	[.42,.46]	[.43,.47]	[.04,.07]
G	0.063	0.443	0.441	0.053
	[.04,.09]	[.42,.47]	[.42,.47]	[.04,.07]
T	0.045	0.423	0.468	0.064
	[.01,.09]	[.35,.49]	[.39,.54]	[.01,.12]

Table 8: Hyperparameter grid search: validation accuracy (%) on fold F0. Selected configuration **bold**. d_e = 8, seq_len = 50, dropout = 0.3 held fixed.

	Hidden size d_h		
LR η	**64**	**128**	**256**
10^{-4}	54.1	55.8	56.2
10^{-3}	57.3	**59.2**	58.9
5×10^{-3}	55.4	57.1	53.8†

† Training diverged in 2/5 runs; mean over stable runs.

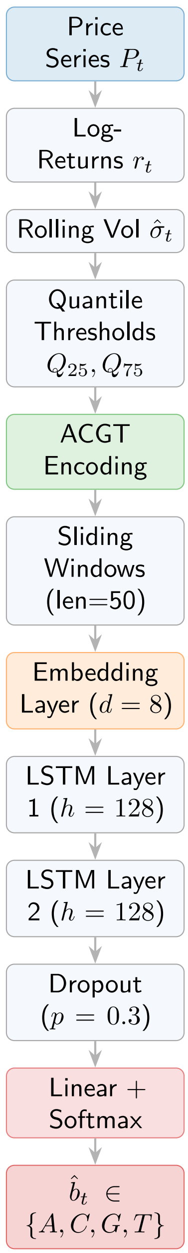

Figure 1: FinancialGenomics pipeline. Blue = input processing, green = discretization, orange = neural encoding, red = prediction.

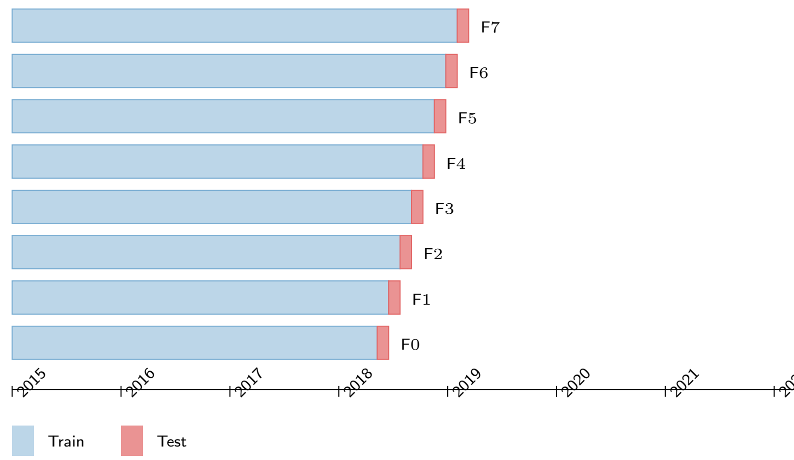

Figure 2: Walk-forward expanding-window validation with 8 folds (F0–F7). Blue = expanding training set; red = held-out test fold (63 days each).

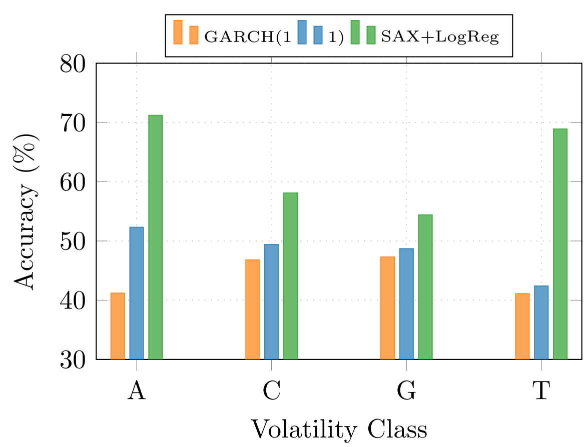

Figure 3: Per-class accuracy by model. Tail classes A and T are most predictable by FinancialGenomics (71.2% and 68.9% respectively), suggesting LSTM captures pre-crash and pre-spike grammar patterns.

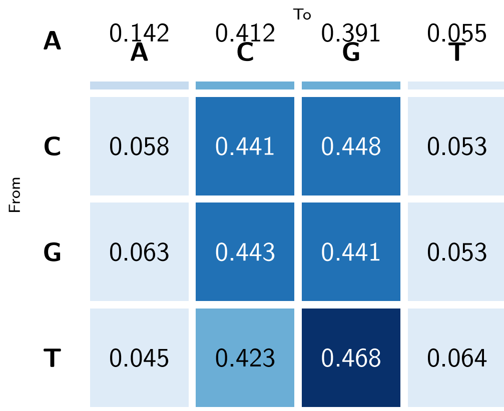

Figure 4: Markov transition matrix estimated from the 2,012-day training corpus. Dark blue = high probability. The C→G and G→C entries dominate (≈0.44), while T→G (0.468) captures the spike-to-growth recovery pattern.

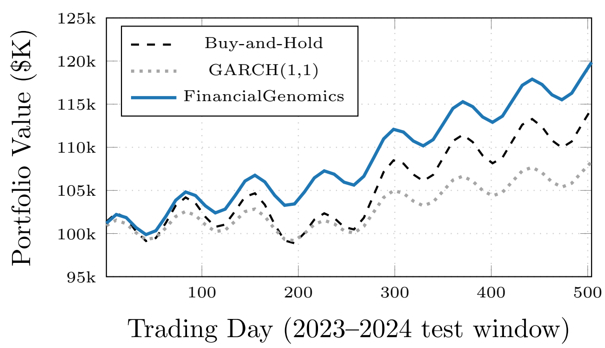

Figure 5: Equity curves on the 504-day test window (2023–2024), initialized at $100,000. FinancialGenomics ends at $118,400 (Sharpe 1.84) versus $112,800 for buy-and-hold (Sharpe 0.71) and $107,200 for GARCH (Sharpe 0.61).